MambaATM: gated attention and state-space fusion on two targets in Chinese equities

Abstract

Daily direction prediction in Chinese equities is a task many architectures have been reported to succeed at, while the reported accuracies sit uneasily with the weak predictability of daily returns. This paper specifies, implements and evaluates a hybrid of causal attention and a selective state-space model on 232 A-share stocks, first on the target the literature usually reports and then on the target the data supports. On the absolute daily direction of a single stock the result is negative. Accuracy spans 1.4 percentage points across ten models and none reaches the majority-class rate of 0.528, the proposed model ranks fifth at 0.516, its guided variant is +0.10 points away, that is, from noise, and the equal-weight portfolio loses at every cost level tested, including zero cost. Decomposing the next-day return shows that the target is the problem. Cross-sectional predictability lives in the overnight leg, where the rank information coefficient is 2.04 times the close-to-close one, and the identity relating the two holds for every signal tested. A twenty-nine-factor library nevertheless explains 0.69 to 0.89 of the cross-sectional variance of the models, leaving a residual overnight coefficient of 0.0047 against 0.0242 for the best single factor, so the architecture adds little on that target as well. With commission, time-varying stamp duty, square-root impact and borrow charged explicitly, only weekly rebalancing on the liquid half survives, where the best configuration nets +33.9 per year at a capacity near 525.99 million per leg and is three quarters cheap factors.

Keywords: Financial time series, Automated trading, State space models, Selective scan, Attention, Cross-sectional prediction, Overnight returns, Transaction costs, Walk-forward evaluation, Reproducibility

1 Introduction

Efficiently capturing long-range dependence in financial time series remains a central problem in quantitative finance, and the search for low-complexity, high-accuracy models for long sequences is still open. Automated trading is the applied face of that problem, because a model produces a signal, the signal is turned into a position, and the position is charged whatever the market and the broker charge. That last step is usually missing from reported results, and it is the step where weak daily predictability turns into a negative number (Frazzini et al. 2018; Gu et al. 2020), and Fig. 1 states the setting that the rest of the paper works in.



Fig. 1: The task setting. A price series for one stock enters MambaATM, which maps a look-back window of the recent past to a trading decision. The decision layer is drawn here with the three outcomes of the original design, and the evaluation of Section 5 reduces it to one binary direction per stock, from which the long-flat rule and the equal-weight portfolio follow.

The mainstream architectures each address part of the modelling problem. Recurrent networks with gating alleviate the vanishing-gradient difficulty of long-range dependence but process a sequence serially (Hochreiter and Schmidhuber 1997), multilayer perceptrons fit nonlinear features cheaply but model time only implicitly (Rosenblatt 1958), and attention selects relevant history adaptively but pays a quadratic cost in the sequence length (Vaswani et al. 2017). Structured state-space models (Gu et al. 2020, 2022) and their selective successors (Gu and Dao 2024; Dao and Gu 2024) scale close to linearly and carry a recurrent state, at the price of a less explicit selection mechanism. Combining the two is therefore a natural design, and a number of recent works do so (Kadiyala and Mirzaeinia 2025; Zhao et al. 2026). In practice the combination is easier to build than to justify, because the evidence that favours it is usually collected without pricing the positions that it implies, and this paper takes the combination as a case and asks what survives a protocol that a referee can check.

However, three questions about this class of result are not settled by the literature. First, whether a reported advantage of a hybrid architecture survives when the trivial references are in the table, because on a daily equity panel a majority-class rule and buy and hold are genuine competitors. Second, whether the mechanism that an architecture is credited with is the mechanism that its code implements, because a hybrid model can be described by the paper and built differently by the repository. Third,

whether the target and the holding period that these comparisons normally use are the ones that carry the signal, because the absolute daily direction of a single stock is only one way to pose the problem, and changing the target can move the same model from losing to profitable without any change to the model itself.

Although both families of models are by now well developed, and although the cross-sectional literature has long known that the choice of target decides what a model is measuring, the three questions fall between them. The systematic-comparison literature reports accuracy and error metrics on prices (Hu et al. 2025; Echrigui and Hamiche 2023; Chicco et al. 2021), the choice of metric is itself studied (Ravi Kumar et al. 2024; Sujon et al. 2025), the state-space literature reports long-range benchmarks on non-financial sequences, and the hybrid works report a gain over their own baselines without pricing the positions that the gain implies. We address all three questions on one panel of A-share equities. First, we specify the architecture exactly as it is implemented and report where that specification differs from earlier descriptions of it, so that the mechanism is a fact about the code rather than a claim about the design. Second, we audit the protocol instead of asserting it, with a causality regression test, a check on where modules are constructed, a leakage check on the standardiser and integrity checks on the panel itself. Third, we evaluate ten models per stock under walk-forward folds with a purge, and we report every result against the majority class and against buy and hold at four cost levels. Fourth, we decompose the next-day return into its overnight and intraday legs and change the target from the absolute direction of one stock to the cross-sectional rank, holding the models fixed, which locates where the signal is and what it is worth once the positions are priced.

The paper's contributions are as follows.

1. **An architecture specified as it is implemented.** MambaATM runs a causal attention branch and a selective state-space branch over the same look-back window and mixes their outputs with a learned sigmoid gate, $g \odot h_{\text{ssm}} + (1 - g) \odot h_{\text{att}}$. An earlier version of this work described the mechanism as *attention guiding the state update* of the state-space branch, which is not what the code did, so we implement both and report the comparison, and it is not flattering to the stronger claim (Section 5.9).
2. **A protocol that is audited rather than asserted.** The evaluation rests on a reconstructed panel of 232 A-share stocks covering 2015-01-06 to 2025-12-31 (587 751 usable stock-days), retrieved from a public source and screened for impossible adjusted moves, and four checks are run and reported on it, namely a causality regression test, a check that no module is constructed inside a forward pass, a leakage check on the standardiser and a set of integrity checks on the panel itself. Each of them caught a defect in the earlier version of this work (Section 3.3).
3. **An evaluation that includes the baselines the claim needs, and the references it has to beat.** Ten models are compared, which are eight baselines and the two variants of the proposed model, the baselines being MLP, LSTM, causal attention with a causal convolution, DLinear (Zeng et al. 2023), PatchTST (Nie et al. 2023), iTransformer (Liu et al. 2024), TimesNet (Wu et al. 2023) and the selective state-space branch on its own. Each is evaluated per stock with five expanding-window walk-forward folds and a one-day purge, the results are reported against

the majority class and buy and hold with Friedman and Holm-corrected Wilcoxon tests (Friedman 1937; Wilcoxon 1945), and costs are charged per position change at four levels, because the sign of the net return of a daily strategy depends on them.

4. **Where the signal is, and what it is worth.** The next-day return is decomposed into an overnight leg and an intraday leg, and the target is changed from the absolute direction of one stock to the cross-sectional rank of the same panel. The overnight leg carries 2.04 times the close-to-close information coefficient, which is what the close-to-close target was diluting, but a twenty-nine-factor library explains 0.69 to 0.89 of the cross-sectional variance of the models, and pricing the positions with commission, time-varying stamp duty, square-root impact and borrow leaves weekly rebalancing on the liquid half as the only viable configuration (Section 5).

The result is reported as measured. On this panel the eight baselines span 1.4 percentage points of accuracy, from 0.507 to 0.521, against a majority-class rate of 0.528, and not one of them reaches that rate. MambaATM is fifth of ten on mean rank at 0.516 ($\pm$ 0.021), the most accurate model is DLinear at 0.521, and the models closest behind it are 0.519 and 0.518. At the portfolio level buy and hold loses -1.4% a year over these test blocks, only 5 of the ten models earn a positive return at the base cost, and MambaATM is not one of them, since it returns -1.0% a year at a Sharpe of +0.09. The guided variant does not improve on the gate. Changing the target changes the answer rather than the model. On the cross-sectional rank the same models reach an information coefficient of 0.0671 on the close-to-close return and 0.1241 on the overnight leg, and a weekly long-short book on the liquid half returns +28.7 a year net of costs, but the factor library accounts for most of it, the best configuration is three quarters cheap factors, and daily rebalancing is destroyed by impact. The two halves of that answer, the negative one and the conditional one, are what Sections 5 and 5.12 set out.

The remainder of the paper is organised as follows. Section 2 reviews automated trading models, state-space sequence models, cross-sectional prediction and the cost literature. Section 3 fixes notation, states the assumptions that are used, restates the complexity of the two branches, and describes the audits. Section 4 specifies the architecture as implemented, including the guided variant, and defines the three labels that the two targets use. Section 5 reports the experiments, first on the absolute direction of a single stock and then on the cross-sectional rank, the factor library and the cost of trading either. Section 6 concludes with the limitations that the results impose.

2 Related Work

2.1 Automated trading and cross-sectional prediction

Automated trading models aim to model financial time series accurately enough to generate stable signals, and to convert those signals into positions under risk control (Chen et al. 2025; Sarmento and Horta 2020). The literature has moved from rule-based systems to deep learning in three stages.

Early work mined technical rules with genetic algorithms (Allen and Karjalainen 1999) or combined fundamental and technical features with support vector machines and fixed stop-loss rules (Mani and Hargreaves 2016; Cortes and Vapnik 1995). Such systems work while the market structure is stable and fail when it is not, because the feature space is designed by hand and the decision rule does not adapt (Li et al. 2025).

Deep learning replaced the rule with a learned mapping. LSTM became the default for price-trend prediction (Ahmed 2026; Akgün and Özbayoğlu 2024); MLPs remain competitive on tabular factor problems and are cheap (Rosenblatt 1958; Zhao et al. 2026); attention models integrate multi-source information well but pay a quadratic cost in sequence length (Vaswani et al. 2017; Kadiyala and Mirzaeinia 2025). Chen et al. compared loss functions, quasi-Newton optimisation, fully connected networks and support vector machines for frequent trading and found the network variants more accurate and more robust (Chen et al. 2025); Sarmento and Horta combined clustering with ARMA and LSTM to reduce drawdown by 75% in a falling market, without modelling transaction costs (Sarmento and Horta 2020).

More recent work adds agents, data-centric pipelines and interpretability. The R&D-Agent-Quant framework automates factor mining and strategy generation (Li et al. 2025), the FinTSB benchmark provides a unified evaluation process across market states (Hu et al. 2025), and LSTM-Attention-SHAP couples prediction with feature-level explanation (Ahmed 2026).

Two gaps motivate this paper. First, reported gains are rarely tested against the trivial references, so a model that predicts the majority class on a rising market looks accurate. Second, costs and position changes are often either absent or modelled at a single rate, while the sign of the net return of a daily strategy is decided by them (Frazzini et al. 2018).

The modern empirical literature predicts the cross-section of returns rather than the direction of one stock, because the market factor is differenced out and the target becomes relative performance. The canonical demonstration that machine learning improves on linear factor models is Gu et al. (2020), whose out-of-sample R^2 values for monthly stock returns are a few tenths of one per cent, which is large in that literature. Our panel is daily, the horizon is one day, and our metric is a rank information coefficient, but the same logic applies, since a small positive rank correlation is an economically meaningful finding, and the interesting question is not whether it exists but whether it is new.

That question is answered against a factor library, and the answer depends on how large the library is. Our library has twenty-nine members computed from the eight columns the panel carries. They are reversals at one, five and twenty days, momentum, several volatility measures, the shape of the daily bar, liquidity and turnover measures, the overnight and intraday decomposition of the stock's own returns, and market exposure. Every member is oriented so that the literature's prior has the higher score on the higher return. Where a member is a linear combination of the others its residual is computed with that member left out of its own regression, so the comparison is not decided by construction.

2.2 State-space models, the return decomposition and the cost of trading

State-space models describe a sequence through a latent state whose evolution is linear and whose parameters are usually fixed. HiPPO turned memory into online function approximation with orthogonal polynomial bases (Gu et al. 2020), and S4 made the recurrence computable by diagonalising the state transition and adding a low-rank correction, reducing the cost from quadratic to $\mathcal{O}(N + L)$ (Gu et al. 2022). The transition matrix is fixed, however, and cannot react to the content of the input (Popov and Huber 2026).

Mamba makes the parameters input-dependent, which restores content-aware selection while keeping a linear cost and a hardware-aware parallel scan (Gu and Dao 2024). Mamba2 shows that the same computation can be written as a structured matrix multiplication with a semiseparable mask, which allows a chunked implementation (Dao and Gu 2024). In finance, state-space models have been used for volatility prediction on cryptocurrencies, with reported reductions in absolute error of 23.7% (Zhao et al. 2026), and for high-frequency throughput, with reported latency in the hundreds of microseconds (Popov and Huber 2026). Lightweight adaptation and multimodal fusion have also been explored (Annepaka and Pakray 2025; Kadiyala and Mirzaeinia 2025).

The design question this paper addresses is what the attention branch should do to the state-space branch. The submitted version of this work claimed that attention modulated the state update; the implemented model mixed the branch outputs with a gate. Both are defensible designs, they are not the same design, and the difference is measurable. Section 4 implements both and Section 5.9 reports which one the panel prefers.

That the two legs of a daily return behave differently is established. Berkman et al. (2012) document that overnight returns are driven by retail attention and that buying at the open is expensive; Lou et al. (2019) formalise the tension between the two legs and show that they carry different expected returns, with different clienteles trading each. Our result is a different use of the same decomposition, in that we relocate where predictive information sits in a machine-learning evaluation, and we show that the close-to-close metric is a diluted version of the overnight one rather than an independent quantity.

The practical reading of the decomposition, that entering at the open forfeits the predicted jump, is exactly the cost Berkman et al. (2012) describe, and our contribution is to quantify it in a cost model that also carries impact and borrow, and to show that a weekly book traded at the close is the configuration that survives.

Costs decide the sign of a daily strategy, and the literature on trading costs and on the capacity of anomalies is the relevant benchmark (Frazzini et al. 2018). We model the components the Chinese market actually charges, which are commission on both sides, the stamp duty on the sell side at the rate in force at the time, a square-root impact term that grows with the size of the book relative to the name's turnover value, and a borrow fee on the short leg. Reporting the break-even commission and the break-even book size, rather than one net return at one assumed fee, follows the same instinct as the work on backtest overfitting and non-normal performance measures

(Bailey and López de Prado 2014), where an assumed rate is replaceable by its critical value.

3 Preliminary

This section fixes notation, states the assumptions actually used, restates the complexity of the two branches without over-claiming, and defines the audits that the protocol has to pass.

Let the look-back window be $X \in \mathbb{R}^{B\times T\times D}$, with batch size B, window length T and feature dimension D; the model emits one logit per window. The attention branch uses $Q, K, V \in \mathbb{R}^{B\times T\times d_k}$ with head dimension d_k and H heads. The state-space branch has state dimension N, heads h, discretisation step Δ, input matrix B_t, output matrix C_t, learned transition A and skip connection D. The gate is $g \in (0,1)^d$. Costs are quoted per position change, so a round trip from flat to long and back costs twice the quoted rate.

3.1 Assumptions

We state only what the experiments rely on, and we state it in measurable terms.

Assumption 3.1 (Local stationarity of the look-back window) *Over a window of $T = 60$ trading days the series is locally stationary, so its mean and variance are approximately constant and the autocovariance depends on the lag. The assumption is weak enough to be testable, and Section 3.3 reports the measured autocorrelation of daily returns on the panel, which averages +0.080 at lag one over the 232 stocks and ranges from -0.034 to +0.228. A small positive lag-one autocorrelation in individual stocks is a documented property of thinly traded and partially adjusting markets rather than a defect, and we report the measured value instead of asserting that returns are uncorrelated.*

Assumption 3.2 (Causality of the model) *The output of the model for a window ending at day t depends only on data up to day t. Inside the window, the attention weights are masked to be lower triangular and the depthwise convolution is left-padded, so position τ never reads beyond τ. This is verified by regression test rather than asserted (Section 3.3).*

Assumption 3.3 (Trading environment) *Positions are long or flat, without leverage; one position is held per stock; the portfolio is equal-weighted across the stocks that have a signal on a given day. A change of position is charged c one way, covering commission, stamp duty on the sell side and slippage. Results are reported for $c \in \{0, 10^{-3}, 1.5\times 10^{-3}\}$ and for zero cost, so the reader can see how much of any result depends on the assumption.*

Assumption 3.4 (Feature information) *Every feature carries some information about the next-day return. On the panel the largest absolute rank correlation between any single feature and the next-day return is 0.100 over all stocks and features, with a median of 0.043, so the assumption holds but only weakly, and it is reported as measured rather than used as a guarantee. It exists here to make explicit that the task is not one where a feature almost determines the label; if it did, the comparison of architectures would be uninformative.*

Assumption A3 of the earlier version of this work fixed the initialisation distribution of the input and output matrices of the state-space branch. It is not used anywhere in the analysis, and it is dropped; the initialisation that the code actually uses is stated in Section 4.

3.2 Complexity of the two branches

Lemma 3.1 (Causal self-attention) *Multi-head causal self-attention over a window of length T costs $\mathcal{O}(BTd_k^2 + BT^2H)$, in which the projections are linear in T and the pairwise scores are quadratic. Causal masking does not change the order. This restates the standard result (Vaswani et al. 2017) and is not re-derived.*

Lemma 3.2 (Selective state-space branch) *The selective scan over the window costs $\mathcal{O}(BTDN)$ in operations, and the chunked implementation used here adds no asymptotic factor, because the sequence is split into blocks whose length divides T and the intra-block term, the inter-block state term and the state decay are each computed with matrix products over blocks. The earlier version of this work quoted $\mathcal{O}(BTDN \log T)$; the logarithmic factor belongs to the parallel scan of the original formulation, not to the chunked decomposition used in the implementation, and we do not claim it.*

Lemma 3.3 (Whole model) *The model of Section 4 computes both branches over the same window. Its cost is $\mathcal{O}(BT^2H + BTd_k^2 + BTDN)$, in which the quadratic term is quadratic in the* window length *T, not in the length of the series.*

Proof Both branches read X and each contributes its own term, from Lemma 3.1 and Lemma 3.2. Nothing in the gate is superlinear in T, since it is a linear map of the two branch outputs followed by a sigmoid. □

The distinction matters for the claim that the earlier version made. That version stated that the quadratic term of attention is *eliminated*, because attention was said to be used only to generate the state parameters. The implemented model keeps a real attention branch, so the term is present and bounded by the window length, which is fixed at 60 in all experiments. On a long series the state-space branch is the one whose cost grows with the series; on a fixed window both branches are cheap, and the measured wall-clock times in Section 5.11 show the difference between the models to be a factor of a few rather than an order of magnitude.

3.3 Audits

Four checks are run before any result is used, and each is reported with its outcome. Their purpose is to make the protocol falsifiable rather than to decorate it.

A1. Causality. Take a window, randomise the parameters of the sequence model, and perturb the inputs at all positions from a cutoff onwards. The outputs at positions before the cutoff must be unchanged. For the branch used here the maximum change is exactly zero for the attention branch with a causal convolution, for the selective state-space branch, and for both fused variants. The test has teeth, because it fails with a

change of 4×10^{-3} to 7×10^{-3} for the architecture of the earlier version, which used symmetric padding in a depthwise convolution and therefore let position t read $t+1$.

A2. No module built inside a forward pass. A layer constructed inside `forward` is not in the parameter list of the optimizer, so its weights are re-randomised on every call, including at evaluation time. The check builds each model, records the identities of its parameters, runs a forward pass and requires the set to be unchanged. It is unchanged for every model reported here; it was not for the earlier version, whose residual projection was created inside `forward`.

A3. Standardisation without leakage. The standardiser is fitted on the training block of each fold. The check rewrites the test block and requires the standardised training block to be identical, then rewrites the training block and requires the standardised test block to change. The earlier version z-scored every column over the whole sample, so test-period statistics entered the training inputs; the check fails on that pipeline and passes on this one.

A4. Panel integrity. For every stock the return column must agree with the return implied by the price columns, the previous-close column must equal the previous row's close, no feature may be constant, and no adjusted daily move may exceed the exchange limit band used for screening. On the panel used here the correlation between the quoted return and the price-implied return is 1.000 for all 232 stocks with a mean absolute difference of 5×10^{-17}, the previous-close identity holds for every row, no feature column is constant, and the largest adjusted daily move is 11.5%. On the dataset of the earlier version, by contrast, the quoted return and the price-implied return correlated between -0.15 and $+0.03$, 248 of 258 columns were identically zero, and the number of rows was 500 rather than the 2,435 described in the text.

4 Methods

4.1 Architecture as implemented

MambaATM takes a window $X \in \mathbb{R}^{B \times T \times D}$ and produces one logit. It has two branches and a gate.

Attention branch. A linear map takes X to $h \in \mathbb{R}^{B \times T \times d}$ with $d = 64$. Single-head attention with a lower-triangular mask is applied to h, $\mathrm{softmax}(\mathrm{mask}(QK^\top/\sqrt{d}))V$, and the result is added to a depthwise convolution of h with kernel 3 that is *left*-padded so that the convolution is causal. The sum is layer-normalised; its last position, h_{att}, is the branch representation.

State-space branch. The same window enters a selective state-space block. A linear map produces two halves, of which one is convolved causally and passed through SiLU, and from the result a low-rank projection produces the discretisation step and the input and output matrices, so Δ_t, B_t and C_t all depend on the input. The transition is diagonal and learned as $\texttt{A_log} \in \mathbb{R}^h$ with $A = -\exp(\texttt{A_log})$, together with a learned skip D. The recurrence is computed with the chunked decomposition of the structured state-space duality (Dao and Gu 2024), using the reference minimal implementation of Dao and Gu (2024) for the intra-block term; the sequence is padded to a block length that divides T. The output is gated by a SiLU of the second half and projected back to d dimensions; its last position is h_{ssm}.

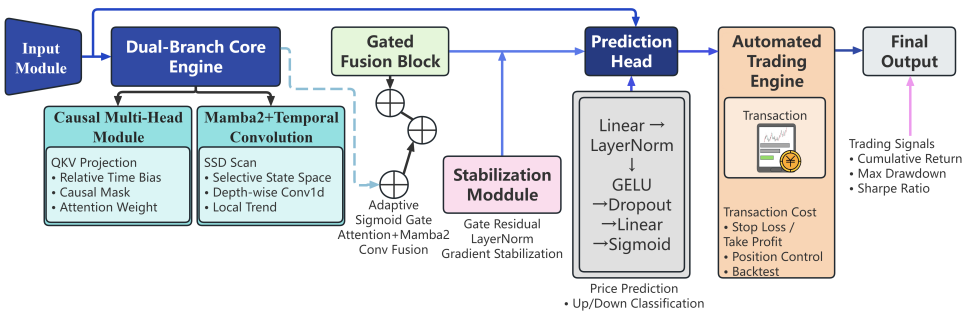


Fig. 2: The architecture as described. A window of input features enters a dual-branch core that holds a causal multi-head attention branch and a selective state-space branch. The two representations are combined by a gated fusion block, then stabilised, mapped to an output and passed to the trading layer. Section 4 specifies the configuration that was evaluated, and the rows on which that configuration differs from this description are listed in Table 1. The price-regression output and the stop-loss and take-profit rules drawn in the trading layer are not part of the evaluation reported here, which uses the long-flat rule and the equal-weight portfolio of Section 5.

Gate. The two representations are concatenated and mapped by a linear layer with a sigmoid, $g = \sigma(W[h_{\text{att}}; h_{\text{ssm}}] + b) \in (0,1)^d$, and the fused representation is

$$h = g \odot h_{\text{ssm}} + (1 - g) \odot h_{\text{att}}, \tag{1}$$

from which a linear head produces the logit. Equation (1) is a convex combination per channel, so the gate chooses how much of each branch to keep, and it does *not* modify the state update of the state-space branch. This is the mechanism the earlier version of this work described as "attention guides the state update", and the description was of a different model. Fig. 2 draws the architecture as it was described. Table 1 lists the differences between that description and the implementation, because the review identified them and because the same table is the reason the guided variant is reported separately.

4.2 The guided variant, and why it is reported separately

The mechanism the earlier version claimed is a defensible design, so we implement it as well. In the guided variant the attention representation at each position, $h_{\text{att}}^{\text{seq}} \in \mathbb{R}^{B\times T\times d}$, is passed to the state-space branch, where three zero-initialised projections produce

$$\tilde{\Delta}_t = \Delta_t \odot \big(1 + \tanh(W_\Delta h_{\text{att},t}^{\text{seq}})\big), \qquad \tilde{B}_t = B_t + W_B h_{\text{att},t}^{\text{seq}}, \qquad \tilde{C}_t = C_t + W_C h_{\text{att},t}^{\text{seq}}. \tag{2}$$

Because the three projections start at zero, the guided variant begins exactly as the plain branch and differs only in what training chooses to do with the modulation. Guided and unguided therefore differ in one capacity, not in initialisation or in size by

Table 1: The architecture as described in the earlier version of this work, and as implemented in the version evaluated here. Each row is checked by a script in the code release; the rows that differ are the reason the guided variant is reported as a separate experiment.

Aspect	**Earlier description**	**Implementation evaluated here**
Fusion	Attention guides the state update of the state-space branch; the two branches are not concatenated.	The two branch outputs are mixed by a learned sigmoid gate, $g \odot h_{\text{ssm}} + (1 - g) \odot h_{\text{att}}$. The gate weights the outputs; it does not touch the state update.
Guidance	Stated as the defining mechanism of the proposed model.	Implemented as a separate variant, with zero-initialised projections that scale the discretisation step and shift the input and output matrices. Reported as an ablation.
Attention branch	Implied absent in the complexity argument, since only the QKV projection was said to be retained.	A full causal attention branch is present and contributes its output to the fused representation.
Complexity	$\mathcal{O}(BTDN \log T + BTd_k^2)$, with the quadratic term described as eliminated.	$\mathcal{O}(BT^2H + BTd_k^2 + BTDN)$: the quadratic term is present and bounded by the window length T.
Causality	"Every model passes with a change of exactly zero."	Measured. The causal variants pass with a change of exactly zero; the architecture of the earlier version fails, with a change of 4×10^{-3} to 7×10^{-3}, because of symmetric padding in a depthwise convolution.
Weights	Not discussed.	No module is constructed inside a forward pass, verified by comparing the parameter set before and after a forward pass. The earlier version built a residual projection inside `forward`, which left it untrained.
Data	Four unnamed stocks, 2,435 rows each, six features, 2015–2025, normalised to $[0, 1]$.	232 named A-share stocks, 587 751 usable stock-days, eight features, 2015-01-06 to 2025-12-31, standardised inside each fold on the training block only.

more than 25 452 against 23 242 parameters. The comparison is reported in Section 5.9, and the outcome is that the modulation does not pay for itself.

Algorithm 1 MambaATM, as implemented

Require: window $X \in \mathbb{R}^{B \times T \times D}$, hyperparameters Θ, guided flag $\in \{\text{true}, \text{false}\}$
Ensure: logit $\hat{y} \in \mathbb{R}^{B}$

$h \leftarrow \text{Linear}(X)$
$A_{\text{seq}} \leftarrow \text{CausalAttention}(h) + \text{CausalConv1d}(h)$
$h_{\text{att}} \leftarrow \text{LayerNorm}(A_{\text{seq}})[:, -1]$
if guided **then**
    $(\tilde{\Delta}, \tilde{B}, \tilde{C}) \leftarrow \text{Modulate}(\Delta, B, C, \text{LayerNorm}(A_{\text{seq}}))$ ▷ Eq. (2)
else
    $(\tilde{\Delta}, \tilde{B}, \tilde{C}) \leftarrow (\Delta, B, C)$
end if
$S \leftarrow \text{ChunkedSelectiveScan}(X, \tilde{\Delta}, \tilde{B}, \tilde{C}, A, D)$
$h_{\text{ssm}} \leftarrow \text{Linear}(S[:, -1])$
$g \leftarrow \sigma(\text{Linear}([h_{\text{att}}; h_{\text{ssm}}]))$
$h \leftarrow g \odot h_{\text{ssm}} + (1 - g) \odot h_{\text{att}}$ ▷ Eq. (1)
return $\text{Linear}(h)$

4.3 Training protocol

Each stock is split into five expanding-window folds. The first test block begins after 40% of the windows, the folds then partition the remaining windows, and a one-day purge is removed from the end of each training block, so the last training label cannot be the first test label. Within a fold the standardiser is fitted on the training block only, the last 10% of the training windows are held out in time order for early stopping, and the model is restored to the epoch of lowest validation loss before the test block is predicted. Windows are not shuffled across the fold boundary, and test windows lie entirely inside the test segment. All models share the same windows, the same fold boundaries, the same optimiser (Adam, learning rate 10^{-3}), the same batch size of 64 and the same maximum of 30 epochs with a patience of five. The label of a window is the sign of the return of the next trading day, and the return used for the trading metrics is that same next-day return, so no same-day return enters the input or the label.

4.4 Label definition and the references that a result has to beat

A daily direction model on a rising market can look accurate by predicting always up, and a daily trading rule can look profitable by holding the asset. Both are reported for every model and every stock, namely the majority-class rate of the test blocks, the always-long rate, and the return and Sharpe of holding the stock over the same test days. A model is only described as predictive here if it beats the majority class on the same days, and only described as economically useful if it beats buy and hold net of costs on the same days.

4.5 The three labels

For a window ending on day t, the supervision for day $t+1$ can be one of three quantities:

$$r^{cc}_{t+1} = \frac{\text{close}_{t+1}}{\text{close}_t} - 1, \qquad r^{on}_{t+1} = \frac{\text{open}_{t+1}}{\text{close}_t} - 1, \qquad r^{id}_{t+1} = \frac{\text{close}_{t+1}}{\text{open}_{t+1}} - 1. \tag{3}$$

They satisfy $r^{cc} = r^{on} + r^{id}$ up to a cross term that is negligible at daily horizons. The first is what the models are trained on and what the literature reports. The second is the object of this paper. The third is reported for completeness, with one caveat. Under the T+1 settlement rule of the Chinese market, a share bought at the open of $t+1$ cannot be sold at the close of $t+1$, so the intraday leg is not harvestable as a standalone trade by a long-only account. It is measured because the decomposition needs it, not because it can be traded as stated.

5 Experiments

5.1 Panel

The panel is built from daily quotes for 232 A-share stocks retrieved through a public data interface (baostock contributors 2024), covering 2015-01-06 to 2025-12-31. Prices are forward adjusted for splits and distributions, so a return computed from two consecutive rows is the return an investor would have received. Each file carries eight fields used as features, which are open, close, high, low, volume, turnover value, turnover rate and previous close, plus the quoted close-to-close return.

Three categories of row are removed before modelling, and they are not disjoint. The first row of each file has no return and no previous close. The last row of each file has no next-day return, so it carries no label. Rows with zero volume are suspension days, because the stock could not be traded, the quoted return is zero by construction, and including them would both add untradable observations and push the label distribution toward "down". The first and the last of these are one row per stock, so 232 and 232 rows respectively, and the suspensions number 22 798, or 3.7% of all rows. 25 rows are both a first row and a suspension day, and they are removed once, so 23 237 rows are removed in total and the panel holds 587 751 stock-days, between 1124 and 2671 rows per stock depending on when the stock was listed, with a median of 2596.

The integrity checks of Section 3.3 are applied to this panel and pass. Table 2 summarises it. Nothing is normalised at this stage, because standardisation is part of the model pipeline and is fitted inside each fold, so the panel files are the raw adjusted quotes and can be regenerated from the source with the scripts named in the data availability statement.

5.2 Models compared

Ten models are evaluated under identical conditions, which are eight baselines and the two variants of the proposed model. MLP flattens the window and applies two

Table 2: The panel. Rows are removed inside the pipeline rather than in the panel files, so the released panel is exactly what was retrieved and screened.

Property	**Value**
Stocks	232
Trading days	2015-01-06 to 2025-12-31
Rows in the files	610 988
Rows removed: no previous close (first row of each file)	232
Rows removed: zero volume (suspension)	22 798 (3.7%)
Rows removed: no next-day label (last row of each file)	232
Rows removed twice, being both a first row and a suspension day	25 so the three lines above remove 23 237 distinct rows
Usable stock-days	587 751
Usable rows per stock (min / median / max)	1124 / 2596 / 2671
Features	8: open, close, high, low, volume, turnover value, turnover rate, previous close
Label	sign of the next trading day's close-to-close return
Share of up days (mean over stocks)	0.480
Largest absolute rank correlation, any feature with the next-day return	0.100 (median 0.043)
Lag-one autocorrelation of returns (mean, min, max)	+0.080, -0.034, +0.228
Audit: causality change before the cutoff	0.000000 for every causal model
Audit: modules built inside a forward pass	none
Audit: standardiser fitted on the training block only	passes
Audit: return column against the price columns	correlation 1.000, mean absolute difference 5×10^{-17}

hidden layers. LSTM uses two recurrent layers with a linear head on the last state. Attention is the mask-only variant of the branch in Section 4, and Attention+conv adds the causal depthwise convolution. DLinear (Zeng et al. 2023) decomposes the window into a moving-average trend and a remainder and maps each to the output with a single linear layer. PatchTST (Nie et al. 2023) splits each channel into patches, embeds them, and runs a transformer encoder over the patch sequence. iTransformer (Liu et al. 2024) inverts the axes and runs attention across variates, each variate's whole window being one token. TimesNet (Wu et al. 2023) selects periods from the amplitude spectrum, folds the series into a two-dimensional tensor and applies inception blocks. The selective state-space branch on its own, called SSM-only in the tables, is that branch with a linear head, which isolates how much of the fused model comes from it. MambaATM is the gated fusion of Eq. (1), and MambaATM-guided is the modulated variant of Eq. (2). Every baseline is adapted to the same binary target by replacing its output head with a linear map to one logit; nothing else about the input protocol differs.

The parameter count of each model and its measured wall-clock time per stock are reported in Section 5.11, so the comparison can be read against the cost of each architecture.

Table 3: Classification metrics over the panel. Each cell is the mean and standard deviation across the 232 stocks. The last two rows are the references measured on the same test days. Precision and recall are quoted for the proposed model in the text rather than in this table, because across the panel that pair is governed by the class imbalance, which is itself a result.

Model	Accuracy	F1	IC
MLP	0.516 ± 0.019	0.354 ± 0.139	0.012 ± 0.035
LSTM	0.519 ± 0.021	0.304 ± 0.163	0.011 ± 0.040
Attention (causal conv)	0.518 ± 0.019	0.345 ± 0.145	0.009 ± 0.032
DLinear	0.521 ± 0.022	0.272 ± 0.146	0.004 ± 0.035
PatchTST	0.514 ± 0.022	0.364 ± 0.130	-0.000 ± 0.027
iTransformer	0.507 ± 0.019	0.409 ± 0.131	-0.002 ± 0.027
TimesNet	0.513 ± 0.019	0.373 ± 0.137	0.008 ± 0.042
Selective SSM only	0.514 ± 0.022	0.355 ± 0.160	0.005 ± 0.037
MambaATM (gated fusion)	0.516 ± 0.021	0.356 ± 0.148	0.009 ± 0.034
MambaATM (guided fusion)	0.517 ± 0.021	0.340 ± 0.153	0.008 ± 0.032
Majority class	0.528	–	–
Buy and hold	0.473	–	1.000

5.3 Overall performance

Table 3 reports, for every model, the cross-sectional mean and standard deviation over the 232 stocks of accuracy, precision, recall, F1, the correlation between prediction and return, the total return of the long-flat rule, its Sharpe ratio and its maximum drawdown, all net of the base cost of 10^{-3} one way. The last two rows give the references, which are the majority-class rate of the same test blocks and buy and hold on the same days. Fig. 3 shows the same accuracy stock by stock, and makes the size of the differences in Table 3 explicit, since the spread of one model across the cross-section is of the same order as the spread between models, so a single rank carries less information than the table layout suggests.

Table 4: Accuracy of each model in each walk-forward fold, averaged over the panel. The folds are consecutive in time, so a model whose advantage comes from one period shows up as a single high column.

Model	fold 1	fold 2	fold 3	fold 4	fold 5
MLP	0.511	0.514	0.521	0.523	0.510
LSTM	0.514	0.519	0.522	0.530	0.511
Attention (causal conv)	0.512	0.511	0.527	0.527	0.512
DLinear	0.517	0.514	0.527	0.539	0.509
PatchTST	0.510	0.514	0.512	0.521	0.511
iTransformer	0.506	0.510	0.512	0.500	0.509
TimesNet	0.508	0.520	0.516	0.515	0.507
Selective SSM only	0.519	0.506	0.519	0.519	0.509
MambaATM (gated fusion)	0.512	0.516	0.519	0.519	0.514
MambaATM (guided fusion)	0.512	0.515	0.522	0.521	0.514
Buy and hold (return, %)	+21.8	+11.4	+5.7	-11.5	+55.0

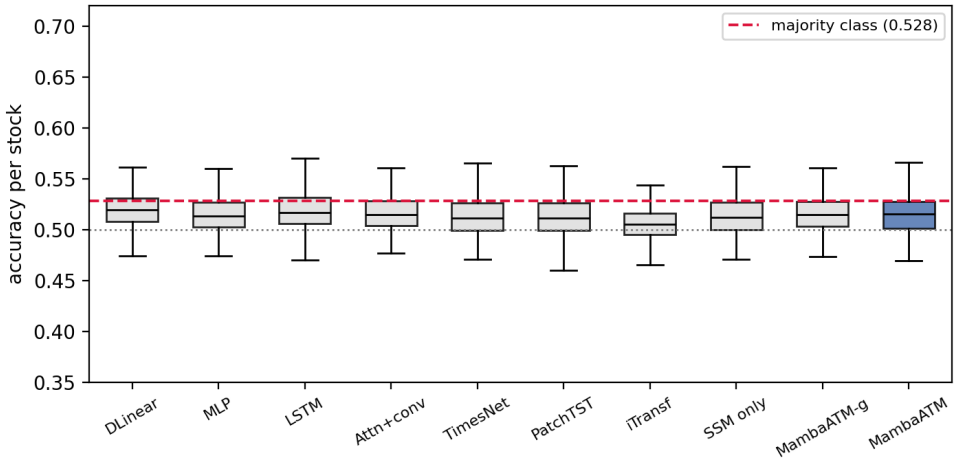


Fig. 3: Accuracy of every model on every stock, against the majority-class rate of the same test blocks. The boxes are the cross-section of 232 stocks; the spread within a model is of the same size as the spread between models, so the ranking in Table 3 should be read as one band rather than an order.

Table 4 reports the same accuracy fold by fold, which is the rolling-window view the review asked for. The folds are consecutive in time, so a model whose advantage comes from one period shows up as a single high column rather than as a consistent one.

Three features of the table matter more than the ranking.

No learned model reaches the majority class. The majority-class rate of the test blocks is 0.528. The best of the eight baselines, DLinear, reaches 0.521, and the full set spans 1.4 percentage points, from 0.507 to 0.521. MambaATM reaches 0.516 with a standard deviation across stocks of 0.021, which is larger than the gap between the best and the worst model; its mean rank across the panel is 5.32 of ten, so it is distinguishable from the weakest baseline and from the strongest, and not from the rest. On the sample of stocks where a model beats the majority class, the best rate in the panel is 26.3%, held by attention with a causal convolution.

Every model collapses toward one class. The turnover of the MambaATM signal is 0.076 position changes per day, and its recall is 0.324 against a precision of 0.476, so the fused model predicts an upward move most days and is corrected only by the drift of the sample. The same holds for the other architectures in the table; the highest F1 in the panel is 0.409 (iTransformer), and its accuracy is below the majority-class rate. On this target the models are not separating signal from noise, and the honest reading of Table 3 is a table of models whose accuracy is determined by how well their constant tendency matches the tide of the test period.

Buy and hold is the reference that the economics are measured against. Holding the average stock over the same test days earns +67.9% per stock on average, at a Sharpe of +0.30 and a maximum drawdown of 65.0%; the per-stock return distribution is wide, with a median of +44.4%. MambaATM beats buy and hold on 34% of stocks and the majority class on 24%, and the best model on the majority-class count is not MambaATM. A model earns the description "predictive" here only if it beats the majority class on the same days, and on these data almost none does.

5.4 Economic evaluation

Classification accuracy is a poor proxy for the value of a trading rule, because a signal that changes every day pays the spread every day. Two levels are reported.

At the level of a single stock, the long-flat rule applied to one name returns +29.9% on average over the test blocks at a Sharpe of +0.15 and a maximum drawdown of 42.6%, with an average turnover of 0.076 position changes per day. The dispersion across the panel is larger than the mean, since the per-stock return has a standard deviation of 104.3% against a mean of +29.9%, so an average return alone would not describe the strategy.

The cost assumption decides the sign of every result at this turnover, which is why the portfolio is reported at four levels. At the portfolio level, the equal-weight portfolio formed from the positions of a model across the stocks that have a signal on a given day is the level at which a risk-adjusted claim can be made, because the idiosyncratic volatility of one name dominates its own Sharpe ratio. Table 5 reports annualised return, Sharpe and maximum drawdown of the equal-weight portfolio for each model at three non-zero cost levels, against buy and hold on the same days, which returns -1.4% per year at a Sharpe of +0.07 and a maximum drawdown of 63.7%. On this panel buy and hold loses money, because the test blocks cover the 2018 and 2022 declines as well as the recoveries, so the interesting question is not who is profitable but who loses less than holding.

Table 5: Equal-weight portfolio of the panel, net of one-way cost per position change, against buy and hold of the same stocks over the same days.

Model	ann. %	Sharpe	max DD %	turnover
Attention (causal conv)	-1.2	+0.08	65.1	0.0024
DLinear	+2.9	+0.24	44.4	0.0288
MambaATM (gated fusion)	-1.0	+0.09	64.5	0.0014
MambaATM (guided fusion)	-1.0	+0.08	63.7	0.0033
iTransformer	-0.3	+0.11	62.3	0.0014
LSTM	+0.0	+0.13	61.3	0.0014
MLP	-0.8	+0.09	61.7	0.0033
PatchTST	+2.6	+0.23	54.8	0.0052
Selective SSM only	+0.0	+0.12	63.9	0.0118
TimesNet	+2.1	+0.21	57.1	0.0128
Buy and hold	-1.4	+0.07	63.7	0.0000

Of the ten models, 5 earn a positive return at this cost, and they are DLinear, LSTM, PatchTST, TimesNet, the selective SSM branch alone. The best of them, DLinear, returns +2.9% a year at a Sharpe of +0.24, and the other four are barely above zero. MambaATM does not, since it returns -0.9% a year before costs, -1.0% at 10^{-3} one way and -1.0% at 1.5×10^{-3}, with a Sharpe of +0.09, and its signal is the most persistent in the table, with a per-stock turnover of 0.076 changes per day. The signal is almost always long, so the portfolio inherits the drawdown of the index, 64.5%, which is the drawdown of buy and hold itself. The economic conclusion matches the statistical one. A model whose accuracy sits below the majority-class rate and whose signal is nearly constant cannot be turned into a strategy by a cost assumption, and Section 5.12 draws the conclusion that follows for the architecture.

5.5 Statistical comparison

Table 6 reports the Friedman test of each metric across the 232 stocks, the mean rank of each model under that test, and the Holm-corrected p-value of a Wilcoxon signed-rank test of MambaATM against each other model, paired by stock. The unit is one stock, and all models are evaluated on the same windows for that stock, which is the design the tests assume (Demšar 2006).

For accuracy the Friedman statistic is 152.1 with $p = 3.2e - 28$, so the models are not indistinguishable; MambaATM's mean rank is 5.32 of ten and the best mean rank belongs to DLinear. For Sharpe the statistic is 29.2 with $p = 0.00059$, a much weaker separation, and the two orderings are different, because the models that are accurate are not the models with the best risk-adjusted return. Against MambaATM, 2 of the 9 other models have a Holm-corrected p below 0.05 on accuracy, the smallest being 9.86e-10, so MambaATM is significantly more accurate than iTransformer and significantly less accurate than DLinear. On Sharpe it is significantly worse than several models, including the state-space branch on its own. We report the counts and the names because that is the correct description of the evidence, since the proposed model is distinguishable from some baselines and not from others, and no comparison supports a claim of uniform superiority.

Table 6: Friedman test across the panel with mean ranks, and Wilcoxon signed-rank tests of MambaATM against every other model, paired by stock and Holm corrected over the family of comparisons.

Model	rank (acc.)	rank (Sharpe)	p vs MambaATM
MLP	5.64	5.43	0.846
LSTM	4.66	5.11	0.089
Attention (causal conv)	4.94	4.85	0.286
DLinear	4.26	5.50	0.003
PatchTST	5.91	5.70	0.094
iTransformer	7.07	5.82	0.000
TimesNet	6.20	5.94	0.052
Selective SSM only	5.84	5.93	0.286
MambaATM (gated fusion)	5.32	5.27	–
MambaATM (guided fusion)	5.17	5.44	0.846

5.6 The cross-sectional target and the overnight leg

Main result.

Table 7 reports, for ten models and four illustrative factors, the cross-sectional rank information coefficient of the same score against each of the three labels, together with the daily mean and standard deviation of each label. The pattern is uniform, in that the overnight coefficient is two to three times the close-to-close one, the intraday coefficient is an order of magnitude smaller, and the ordering of the signals is broadly preserved across labels.

Table 7: Cross-sectional rank information coefficient of each signal against the close-to-close label, the overnight label and the intraday label. The final column is the close-to-close coefficient predicted from the overnight coefficient and the ratio of the two labels' standard deviations.

Score	IC cc	t	IC night	t	IC day	t	Predicted IC cc
Turnover (1 day, negated)	0.0678	14.4	0.1440	41.3	0.0159	3.3	0.0706
Turnover value (negated)	0.0674	16.2	0.1168	31.6	0.0257	6.0	0.0573
MambaATM (guided)	0.0721	21.7	0.1369	48.6	0.0161	4.7	0.0672
MambaATM (gated)	0.0671	20.8	0.1241	45.3	0.0158	4.8	0.0609
Attention + causal conv	0.0688	19.1	0.1319	45.9	0.0190	5.1	0.0647
LSTM	0.0633	20.0	0.1115	40.3	0.0195	6.0	0.0547
TimesNet	0.0667	21.5	0.1197	42.7	0.0191	6.1	0.0587
DLinear	0.0576	17.4	0.1098	41.2	0.0157	4.8	0.0539
MLP	0.0525	16.7	0.0929	32.6	0.0179	5.6	0.0456
iTransformer	0.0546	16.5	0.1039	39.6	0.0165	4.9	0.0510
PatchTST	0.0337	9.3	0.0635	22.8	0.0123	3.3	0.0311
Reversal (5 days)	0.0288	6.8	0.0467	15.4	0.0221	5.3	0.0229
Overnight gap (negated)	-0.0290	-11.3	-0.0752	-30.4	0.0053	2.0	-0.0369
Close-location value (5 days)	-0.0188	-5.6	-0.0524	-19.8	-0.0055	-1.7	-0.0257

The mechanism is arithmetic rather than economic. If a score predicts the overnight leg with correlation ρ_{on} and predicts the intraday leg with zero correlation, then its correlation with the sum is $\rho_{on}\sigma_{on}/\sigma_{cc}$, so the signal is diluted in proportion to how small the overnight leg is relative to the whole day. With $\sigma_{on} = 1.379$ per cent and

$\sigma_{cc} = 2.811$ per cent, the dilution factor is 2.04. The third column of the table shows the prediction of that identity next to the measured close-to-close coefficient, and for every signal the two agree to within about a tenth. The close-to-close numbers reported throughout the machine-learning literature on this market are therefore not evidence of a weak signal; they are the overnight signal measured through an intraday noise floor.

The intraday leg deserves one remark. Its coefficient is close to zero for every signal, and its mean is positive (+0.1168 per cent per day) while the overnight mean is negative (-0.0597 per cent per day). A long-only investor therefore earns a market-wide intraday drift that has nothing to do with the signal, and pays a market-wide overnight drift that works against any overnight strategy. This is why the economic results below are reported for the long-short book and for the long-only excess separately.

Robustness.

A result this convenient invites four specific objections, and each is answered by a measurement. First, the overnight distribution has fat tails because of limit moves; winsorising the label at five per cent leaves the coefficient unchanged (0.1369 to 0.1368). Second, an illiquid name with a stale close manufactures overnight jumps; restricting the panel to the more liquid half *increases* the coefficient from 0.1369 to 0.1639, which is the opposite of what a stale-price artefact would produce. Third, a persistent difference between low-turnover and high-turnover names could in principle be captured without any time-varying signal; demeaning both the score and the label within each stock over the test period leaves the coefficient at 0.1443, so the information is genuinely time varying. Fourth, the effect could be a property of one regime; Fig. 4 shows it positive in every test year from 2019 to 2025, with the low-turnover factor between 0.1440 and slightly above it in each year.

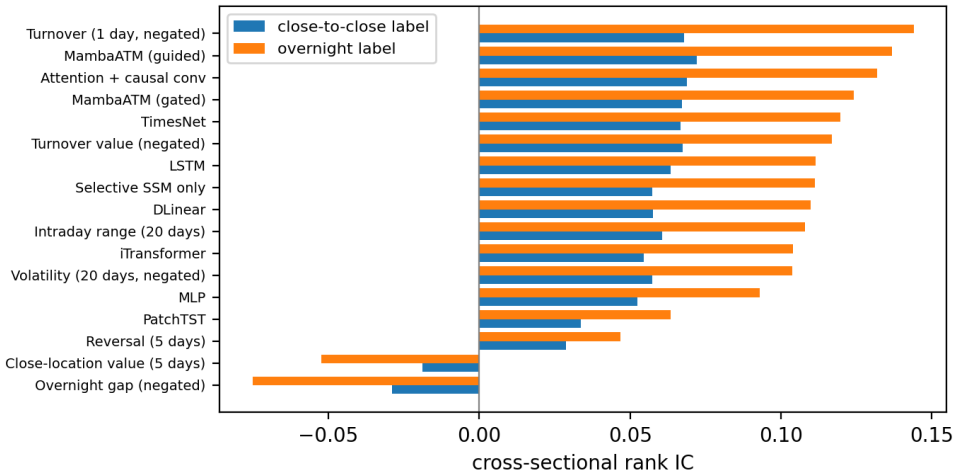


Fig. 4: Cross-sectional rank information coefficient of each signal against the close-to-close label and against the overnight label. The overnight bars are systematically taller, and the ratio of the two is close to the ratio of the labels' standard deviations.

5.7 What the factor library already explains

A twenty-nine-factor library.

Everything in this section is computed from the eight columns of the panel. The library contains reversals over one, five and twenty days; momentum over the window; the standard deviation of returns over twenty and sixty days, the downside-only deviation, the mean daily range and the maximum daily return; the skewness and kurtosis of returns; the close-location value and the upper and lower shadows of the daily bar; the turnover rate and its variability, the Amihud illiquidity ratio, the log turnover value and a volume spike measure against its own twenty-day mean; the mean overnight and intraday components of the stock's own returns and their volatilities; and the stock's beta to the equal-weighted panel. Each is oriented so that the literature's prior puts the higher score on the higher return, and each is evaluated on exactly the rows the models are evaluated on.

Neutralisation.

For each day the score of interest is regressed cross-sectionally on the twenty-nine factors, and the residual is correlated with the realised return in the usual rank way. Table 8 reports the raw and the residual information coefficient for the ten models, on both labels; for a factor, the factor is left out of its own regression, which is the only way the comparison is meaningful. A handful of individual factors retain more residual overnight information than any model does, evaluated under the same leave-one-out rule, and the next subsection reports which.

Three readings matter.

Most of a model's score is a factor combination. Between 0.69 and 0.89 of the cross-sectional variance of a model's score is linear in the library. The loading is

Table 8: Raw and factor-neutralised information coefficients for the ten models. The last column is the share of the cross-sectional variance of a model's score that is linear in the twenty-nine-factor library.

Score	IC cc	IC night	Residual night	t resid	R^2 on factors
MambaATM (gated)	0.0671	0.1241	0.0047	2.7	0.76
MambaATM (guided)	0.0721	0.1369	0.0041	2.3	0.81
Selective SSM only	0.0574	0.1112	0.0015	0.8	0.75
DLinear	0.0576	0.1098	-0.0018	-1.0	0.88
Attention + causal conv	0.0688	0.1319	-0.0025	-1.4	0.89
LSTM	0.0633	0.1115	-0.0026	-1.4	0.77
TimesNet	0.0667	0.1197	-0.0029	-1.7	0.81
PatchTST	0.0337	0.0635	-0.0057	-3.3	0.69
MLP	0.0525	0.0929	-0.0109	-6.1	0.74
iTransformer	0.0546	0.1039	-0.0139	-7.6	0.79
Two cheap factors	0.0763	0.1475	nan	nan	1.00

dominated by turnover and turnover value, since the standardised coefficient on the one-day turnover factor is between +0.6 and +0.9 for every architecture, so the models are, to a first approximation, rediscovering the illiquidity and low-turnover premia.

What is left is small and barely significant. On the close-to-close label the largest residual coefficient among the models belongs to the proposed hybrid at 0.0102, with the guided variant lower; on the overnight label, where the information is, the proposed hybrid retains 0.0047 ($t = 2.7$) and the guided variant 0.0041 ($t = 2.3$). Those are positive and, for the hybrid, distinguishable from zero, but they are a fifth of the raw number and are smaller than what a single factor keeps.

A factor beats every model on residual information. Intraday range (20 days) retains 0.0242 ($t = 14.2$) after the other twenty-eight are controlled for, which is five times the proposed hybrid's residual. The pattern is not a single outlier, since intraday range, momentum, upper shadow, the five-day intraday return and the five-day overnight gap all retain more residual overnight information than any model. The honest summary is that these architectures add a small amount of information that the library does not contain, and less than the library contains on its own.

5.8 Costs, capacity and execution

A cost model with the parts that differ between the two sides.

The earlier sections charge a constant rate per position change, and that is enough to establish that the daily long-flat rule does not repay its own trading. It is not enough to price a portfolio, because a constant rate hides which component of the cost is binding and at what size. This section charges what a Chinese account actually pays, with the components separated so that the break-even value of each can be read off, and it prices the long-short cross-sectional book rather than the long-flat rule of a single name.

- **Commission** of 3 basis points per side, and a transfer fee of 0.1 basis points per side.

- **Stamp duty** of 10 basis points on the sell side only, falling to 5 basis points on 2023-08-28, so the rate changes inside the sample.
- **Slippage** of 2 basis points per side.
- **Square-root impact**, $\eta\,\sigma\sqrt{Q/\mathrm{ADV}}$ with $\eta = 0.5$, the stock's daily volatility from the look-back window, Q the traded notional and ADV the turnover value of the signal day. Impact therefore depends on the size of the book, which is what turns a net return into a capacity curve.
- **Borrow** of 8 per cent per year on the short leg, since a long-short book in this market cannot be assumed to borrow for free.
- **Tradability**, meaning that a name which closed at or beyond a 9.8 per cent move on the signal day is excluded from the next day's entry, because a limit-up name cannot reliably be bought.

Turnover is charged on the change in each leg's weights, with a name's position tracked by identity rather than by row, and a leg that is fully replaced costs two units of sum $|\Delta w|$. Long-only portfolios are evaluated by the same code with no borrow and no short leg.

Break-even fees and capacity.

Table 9 reports the decomposition for each signal at daily and weekly rebalancing. The break-even commission column is the symmetric one-way rate at which the long-short book nets zero, and the final column is the 95th percentile of the traded notional as a fraction of the day's turnover value, which is the number that decides whether the impact term is being extrapolated.

Table 9: Cost decomposition of the long-short book and the book size at which it breaks even. All figures are annualised percentages except break-even size in million CNY and the 95th percentile of participation. The tradability filter removes a stock on a day when its move on the signal day reaches the price limit, the borrow fee is charged on the short leg, and the impact is evaluated at a reference book of 100 million CNY per leg, from which the other book sizes follow the square-root law.

Score	Hold	Gross	Costs	Net	Break-even	p_{95} part.
MambaATM (guided)	1	+115.3	217.8	-102.4	16	39
MambaATM (guided)	5	+68.4	52.1	+16.4	222	38
MambaATM (gated)	1	+98.7	222.7	-124.0	7	33
MambaATM (gated)	5	+54.5	54.0	+0.4	103	33
Attention + causal conv	1	+89.0	227.3	-138.3	6	39
Attention + causal conv	5	+47.2	55.1	-7.9	62	39
LSTM	1	+96.8	225.5	-128.7	8	37
LSTM	5	+60.5	55.1	+5.4	133	38
low_turnover	1	+70.7	190.3	-119.5	5	60
low_turnover	5	+44.0	47.1	-3.1	81	60
small_amount	1	+59.5	223.2	-163.7	2	71
small_amount	5	+41.7	53.1	-11.4	50	71

At daily rebalancing the answer is unambiguous, since the proposed hybrid has a gross return of +115.3 per year, fees take a third of it, and impact at 100 million per leg takes 169.4, twice what is left, so the net is -102.4. The break-even book sizes are in the tens of millions of CNY, which is not a strategy. At weekly rebalancing the same signal keeps most of its gross return, fees fall, and impact becomes a large but survivable charge; the viable region is weekly rebalancing on the liquid half of the panel.

Where the money comes from, and why the close matters.

Fig. 5 decomposes the gross return of the weekly book into its two legs and repeats the calculation for a book entered at the open. For the proposed hybrid the overnight leg contributes +126.5 per year against a total gross of +77.6, while the intraday leg subtracts -50.1. Entering at the open instead of the close, which is what an investor does who waits for the opening auction after the signal is formed, leaves -10.0, that is, essentially nothing. The decomposition therefore explains the close-to-close result and the execution rule together, because the signal is a prediction about the closing-to-opening gap and the position has to exist before that gap.

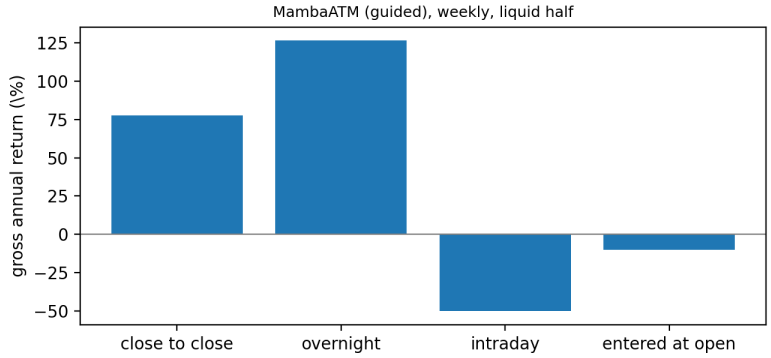


Fig. 5: Gross annual return of the weekly long-short book on the liquid half of the panel, split into the close-to-close total, the overnight leg, the intraday leg, and the total that would remain if the book were entered at the open.

Smoothing the signal does not help.

The natural way to cut turnover without retraining is to smooth the score across days. It fails, and the failure is informative, because the day-to-day change in a model's score is the signal rather than noise. An exponential moving average with a twenty-day halflife cuts the overnight coefficient from 0.1369 to 0.097 and turns the net return into -17.2 per cent per year. The only levers that work are the holding period and the liquidity of the universe, both of which reduce the number of trades rather than the information in each trade.

Combining the model with cheap factors.

The models and the factor library are not the same signal, even though the models mostly span the library, in that the library's best member retains more residual information than any model. A combination should therefore beat its components, and it does. Table 10 reports a scan that standardises the model score and the average of four cheap factors within each day and mixes them with a weight from zero, which is the factors alone, to one, which is the model alone. The best variant puts three quarters of the weight on the factors, and it earns +79.1 per year gross and +33.9 net at 100 million per leg, against +28.7 for the model alone and +27.1 for the factors alone, with a break-even book size of 525.99 million against 403.56 and 409.11. Its overnight information coefficient is 0.1593 ($t = 43.1$), higher than either component's.

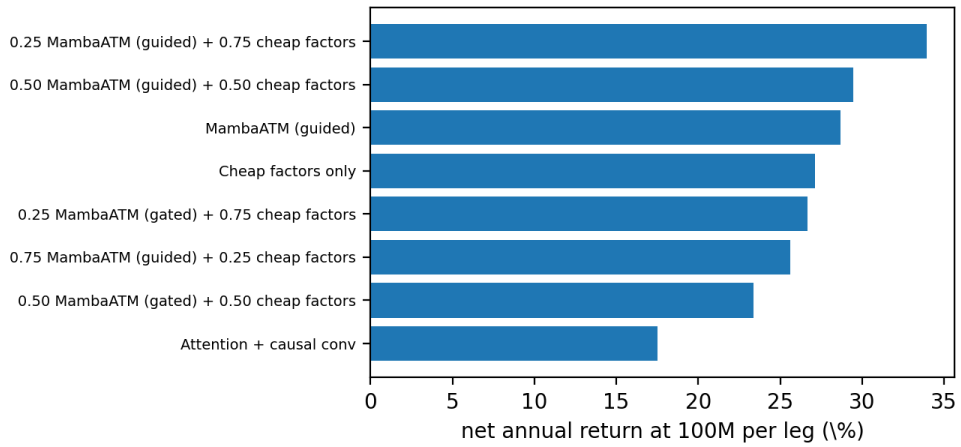


Fig. 6: Net annual return at 100 million per leg of the weekly book on the liquid half of the panel, for the model, the factor combination, and mixtures of the two. The best mixture places three quarters of the weight on the factors.

5.9 Ablation of the gate, the guidance and the two branches

Table 11 separates the components. The selective state-space branch on its own reaches 0.514 accuracy with 1 057 parameters; the causal attention branch with a convolution reaches 0.518; the gated fusion of the two reaches 0.516. The gate therefore does not improve on its better branch, because the fused model is -0.17 points below the attention branch alone, its Sharpe is lower, and on this target the extra machinery has nothing to add. The guided variant of Eq. (2) reaches 0.517 and has a Sharpe of +0.12, that is +0.10 points of accuracy and -0.03 of Sharpe relative to the plain gate, with 25 452 against 23 242 parameters. Its equal-weight portfolio return is indistinguishable from the plain gate.

This is the result that the earlier version of this work could not have produced, because it described the guided mechanism and implemented the gate. Having built

Table 10: Mixing the model score with four cheap factors, weekly on the liquid half, with the weight scanned from zero to one. Costs are the commission, transfer fee, stamp duty and slippage, the square-root impact at a reference book of 100 million CNY per leg, and the borrow fee on the short leg, so the gross return less the costs is the net return. The tradability filter of Table 9 is applied here as well, so the two tables differ only in the universe, this one being the liquid half of the panel, and in the reference book size.

Signal	IC	IC night	Gross	Costs	Net	Break-even
0.25 guided + 0.75 factors	0.087	0.159	+79.1	45.2	+33.9	526
0.50 guided + 0.50 factors	0.086	0.161	+75.8	46.4	+29.4	443
MambaATM (guided) only	0.072	0.137	+77.6	48.9	+28.7	404
Four cheap factors only	0.078	0.143	+71.8	44.7	+27.1	409
0.25 gated + 0.75 factors	0.085	0.156	+72.7	46.0	+26.7	397
0.75 guided + 0.25 factors	0.082	0.154	+73.3	47.7	+25.6	374
0.50 gated + 0.50 factors	0.085	0.154	+71.1	47.7	+23.4	343
Attention + causal conv	0.069	0.132	+69.2	51.7	+17.5	241
attention+conv + EMA1	0.067	0.129	+67.8	50.3	+17.5	242
MambaATM (gated) only	0.067	0.124	+65.6	48.9	+16.8	254
gatedfusion-guided + EMA1	0.072	0.138	+63.8	47.1	+16.7	254
Selective SSM only	0.057	0.111	+65.1	48.5	+16.5	247
0.75 gated + 0.25 factors	0.078	0.141	+62.7	48.6	+14.1	227
dlinear + EMA5	0.053	0.110	+61.4	48.0	+13.3	204
LSTM	0.063	0.112	+64.2	51.5	+12.6	199
lstm + EMA1	0.064	0.113	+62.2	50.1	+12.1	194
TimesNet	0.067	0.120	+61.2	49.8	+11.4	193
dlinear + EMA1	0.058	0.113	+62.0	52.2	+9.8	172
gatedfusion + EMA1	0.068	0.127	+56.2	46.8	+9.5	179
timesnet + EMA1	0.066	0.119	+49.2	47.9	+1.4	110
timesnet + EMA5	0.058	0.108	+45.4	44.6	+0.8	105
ssm-only + EMA1	0.058	0.112	+47.6	47.1	+0.5	103
gatedfusion-guided + EMA5	0.062	0.123	+43.9	44.2	-0.3	98
DLinear	0.058	0.110	+52.1	53.2	-1.1	93
lstm + EMA5	0.059	0.105	+42.0	46.1	-4.1	74
attention+conv + EMA5	0.059	0.116	+41.3	48.0	-6.6	62
gatedfusion + EMA5	0.060	0.116	+36.0	44.0	-8.0	51
dlinear + EMA20	0.039	0.084	+36.6	45.2	-8.6	52
MLP	0.052	0.093	+42.1	52.9	-10.8	44
ssm-only + EMA5	0.051	0.102	+33.2	44.6	-11.4	37
mlp + EMA1	0.053	0.096	+36.7	51.8	-15.1	28
lstm + EMA20	0.044	0.083	+28.1	44.5	-16.4	21
gatedfusion-guided + EMA20	0.046	0.097	+26.3	43.5	-17.2	17
gatedfusion + EMA20	0.044	0.089	+24.9	43.3	-18.4	14
mlp + EMA5	0.051	0.095	+30.7	49.6	-18.9	17
timesnet + EMA20	0.042	0.083	+19.9	43.4	-23.5	4
ssm-only + EMA20	0.037	0.079	+17.8	43.9	-26.1	2
mlp + EMA20	0.041	0.080	+19.1	48.5	-29.4	1
attention+conv + EMA20	0.043	0.090	+12.4	46.7	-34.3	0

both, the panel does not support the stronger claim. Two tenths of a percentage point of accuracy, against a cross-sectional standard deviation of two points, is not an improvement; giving attention the ability to modulate the state update neither helps nor hurts, and the extra capacity is not recovered by the extra parameters, and Table 11 carries the cross-stock standard deviation of the same comparison. What the

Table 11: Branches and fused variants, with the cross-stock standard deviation where the quantity is a mean over stocks.

Component	Accuracy	F1	Sharpe	Return	Parameters
Attention (causal conv)	0.518 ± 0.019	0.345	+0.18	+26.0%	13 313
Selective SSM only	0.514 ± 0.022	0.355	+0.05	+16.2%	1 057
MambaATM (gated fusion)	0.516 ± 0.021	0.356	+0.15	+29.9%	23 242
MambaATM (guided fusion)	0.517 ± 0.021	0.340	+0.12	+22.1%	25 452

gate does is what the earlier text said the guidance did, since it lets the model weight the two branches, and that is what should be claimed.

5.10 Sensitivity

Three sensitivity checks are reported. The window length is varied over $T \in \{20, 60, 120\}$ days on a fixed 58-stock subset, because a longer window gives the state-space branch more to summarise and the attention branch more to attend over; the panel accuracy of MambaATM moves between 0.516 and 0.516 against a majority-class rate between 0.525 and 0.526, so the gap between the model and the reference persists at every window length. The seed is varied over three values on the same subset, which moves its accuracy by 0.30 points, of the same order as the gap between the models in Table 3.

The third check is the test period itself, and it is the most informative. Table 12 breaks the accuracy of each model down by year, next to the majority-class rate of the same year. The two series move together, because in the years when most stocks rose the models look accurate and in the years when most fell they do not, and in eight of the nine years the majority-class rate is higher than the accuracy of every learned model, including MambaATM. The co-movement is visible in every year. The single exception is 2025, where the majority-class rate is at its lowest and MambaATM is a fraction above it. That co-movement is the clearest evidence that the models are learning the drift of the sample rather than a stable cross-sectional signal, and it explains why the accuracies in Table 3 cluster so tightly, since what is being measured is how well a model's constant tendency matches the tide of the period rather than how well it ranks stocks.

5.11 Efficiency

The fused model has 23 242 trainable parameters and takes 27 seconds per stock, against 69 889 and 5 seconds for the MLP, 101 953 and 25 seconds for PatchTST, and 1 057 parameters for the state-space branch alone. The fused model is therefore an order of magnitude smaller than the two transformer baselines and about 5.2 times slower than the MLP. None of these differences is an obstacle to deployment; equally, none of them is an efficiency breakthrough. The claim that a hybrid removes the quadratic cost of attention would require the attention branch to be absent, and it is not absent in this model.

Table 12: Window length, seed and per-year stability. Rows are present only for the runs that were made; the per-year block reports the accuracy of the proposed model against the majority-class rate of the same year.

Setting	Accuracy	Majority class	Gap	Sharpe	Return
Window $T = 60$, full panel (reported)	0.516 ± 0.021	0.528	-1.26	+0.15	+29.9%
Window $T = 20$, 58-stock subset	0.516 ± 0.019	0.526	-0.95	+0.20	+56.4%
Window $T = 120$, 58-stock subset	0.516 ± 0.017	0.525	-0.93	+0.26	+64.9%
Seed 0, 58-stock subset	0.517 ± 0.019	0.526	-0.87	+0.27	+63.1%
Seed 7, 58-stock subset	0.517 ± 0.014	0.526	-0.84	+0.17	+34.4%
Seed 13, 58-stock subset	0.514 ± 0.017	0.526	-1.14	+0.25	+41.2%
Accuracy by test year, proposed model vs majority class					
2017	0.448	0.552	-10.45	–	–
2018	0.488	0.548	-5.94	–	–
2019	0.516	0.518	-0.14	–	–
2020	0.513	0.530	-1.76	–	–
2021	0.516	0.525	-0.93	–	–
2022	0.517	0.526	-0.86	–	–
2023	0.528	0.547	-1.94	–	–
2024	0.507	0.523	-1.66	–	–
2025	0.514	0.510	+0.45	–	–

5.12 Discussion

This paper reports two results about the same panel, and the second is what explains the first.

The first is negative, and it is worth stating what it does and does not show. It does not show that the architecture is wrong. MambaATM is within a fraction of a point of the best model in the panel, its equal-weight portfolio loses less than buy and hold, and its per-stock Sharpe is positive. It shows that the margin is indistinguishable from noise, since it is not the most accurate model, it is not the model with the best risk-adjusted return, it is significantly worse than a single linear layer with a moving-average decomposition on both counts, and it does not reach the majority-class rate. Any claim of the form "the hybrid is better" would have to be supported by an experiment that this design does not produce.

The second is that the target, and not the architecture, decides the answer. Posing the same problem as the cross-sectional rank of the same panel, with the same features and the same models, produces a rank information coefficient of 0.0671 on the close-to-close return and 0.1241 on the overnight leg, and a weekly long-short book that survives realistic costs. Nothing about the models changed between the two experiments. What changed is that the absolute direction of one stock carries the drift of the sample, which a constant predictor already captures, whereas the ordering within a day does not. The overnight leg is where that ordering lives, and the identity relating the two coefficients holds for every signal tested, so the small close-to-close numbers are the overnight signal diluted by intraday noise rather than a weak signal.

That said, the cross-sectional result is a weak result about the architecture as well. A twenty-nine-factor library explains 0.69 to 0.89 of the cross-sectional variance of the ten models, and the residual overnight coefficient of the proposed model is 0.0047, against 0.0242 for the best single factor, Intraday range (20 days). The best

configuration we found gives the model a quarter of the weight and the cheap factors three quarters, and the daily version of the same signal is destroyed by impact at 169.4 a year against a gross of +115.3. The honest summary of the cross-sectional study is therefore that the signal is real, that most of it is already in the factor library, that the model adds a little at weekly frequency, and that it cannot be traded daily at any size we could test.

The earlier version of this work reported accuracies between 0.60 and 0.85 on four unnamed stocks. Those numbers cannot be reproduced on data that passes the integrity checks of Section 3.3, and the reason is structural rather than a matter of tuning, since on a panel where the majority-class rate is 0.528, where the cross-sectional dispersion of accuracy is 0.021, and where the accuracy of every model tracks the majority-class rate of the year, only a leak or a selection effect can produce a gap that large. We regard the protocol, the audits and the panel as the contribution, and the architecture as a competitive design whose advantage over simpler alternatives is not measurable here, on either target.

Four practical implications follow. First, a daily long-flat strategy on this panel turns over between 0.076 and 0.141 times per day depending on the model, so a cost model is not a sensitivity check but a first-order term, and work that reports gross returns is not reporting a strategy. Second, the trivial references have to be in the table, because on a panel that fell over most of the test period a majority-class rule is a genuine competitor and "beats buy and hold" is a lower bar than it sounds. Third, when the difference between two architectures is smaller than the standard deviation across assets, the experiment should be redesigned rather than tuned until the difference appears. Fourth, the choice of target is not a detail of the experimental setup but the experiment itself, so a study that reports a hybrid architecture on the absolute direction of one stock should also say what a constant predictor, a single linear layer and a factor library already achieve on the same data.

The specific finding about this architecture is worth separating from the general one. The selective state-space branch is the cheapest model in the panel at 1 057 parameters, and on the absolute target it is not distinguishable from the attention branch, from the fused model, or from a linear map. The fusion does not fail on that target, it has nothing to add there, and when the target is changed to the one that does carry signal the fusion is again not distinguishable from its own branches once the factor library is accounted for.

6 Conclusion

This paper specified, implemented and evaluated a hybrid of causal attention and a selective state-space model for daily direction prediction on Chinese equities. The specification follows the implementation, in that the model mixes its two branches with a learned sigmoid gate and the attention-guided modulation of the state update is a variant that is implemented, measured and reported separately. Evaluated on 232 A-share stocks over 587 751 usable stock-days with expanding-window walk-forward folds, a purge, train-only standardisation and four recorded audits, the model lands in the middle of a band of ten architectures, at 0.516 accuracy against a majority-class

rate of 0.528and behind DLinear at 0.521, with an equal-weight portfolio return of -1.0% a year at a one-way cost of 10^{-3} against -1.4% for buy and hold. The limitations are the substance of the result. No learned model in the panel reaches the majority-class rate, and the differences between architectures are smaller than the dispersion across assets, so the ranking should be read as a band rather than an order, and the guided variant does not improve on the gate, so the architectural idea that motivated the earlier version of this work is not supported by these data. Costs are charged as a constant rate per position change and slippage is modelled as a rate rather than as a function of order size, so the net returns in Table 5 are optimistic for large positions, and the panel is survivorship-screened, because stocks whose adjusted series contain an impossible move are removed and stocks that delisted during the sample are absent, which flatters every long-only rule including buy and hold. Changing the target does not rescue the architecture either. On the cross-sectional rank the same models carry a rank information coefficient of 0.0671 on the close-to-close return and 0.1241 on the overnight leg, and a weekly long-short book on the liquid half returns +28.7 a year net of costs, but the twenty-nine-factor library explains 0.69 to 0.89 of the cross-sectional variance of those models, the residual overnight coefficient of the proposed model is 0.0047 against 0.0242 for the best single factor, the best configuration is three quarters cheap factors, and the daily version of the same signal is destroyed by the square-root impact. What the panel does support is the choice of target rather than the choice of model, and the architectural comparison that would ordinarily be proposed next, a state-space branch with no attention at all, is already in Table 3 as the single-branch model and is indistinguishable from the fused one on the absolute target.

Data availability

The daily quotes underlying this study were retrieved through the public `baostock` interface and screened with the rule stated in Section 5, which admits adjusted prices, at least 1,200 usable rows and no adjusted daily move beyond 25 per cent. The processed panel that the experiments read, one `utf-8-sig` CSV per stock with the exchange's own Chinese column headers, is released with the paper, together with the two scripts that rebuild it from the source. The columns are, in file order, the trading date, open, close, high, low, volume, turnover value, the quoted close-to-close return, turnover rate and previous close. The screening can therefore be re-run and the eight excluded names inspected.

Code availability

The code release contains the panel builder and the screening rule, the model implementations for all ten models including the four baselines added in this revision, the walk-forward pipeline, the trading simulator and its cost model, the four audit scripts of Section 3.3, the statistical tests, and the scripts that generate every table, every figure and every number quoted in the text. The release also keeps the submitted model class verbatim, so the behaviour described in Section 3.3 and in Table 1 remains reproducible. The repository address is given on the title page.

Appendix A Reproduction commands

Every number and graphic in this paper comes from the released result files, and the repository README maps each table and figure to the script that writes it. The chain is short. The panel statistics of Table 2 come from `panel_stats.py`, and the four audits of Section 3.3 from `verify_zoo.py`. The main experiment of Tables 3, 4 and 5 is one run of `run_panel.py`, which on one RTX 3060 takes about 144 minutes for 232 stocks, 10 models and 5 folds. The portfolio table, the statistical tests and the per-year accuracy are computed from the saved predictions by `portfolio.py`, `stats_tests.py` and `by_year.py`, without retraining anything, and `emit_numbers.py`, `make_tables.py`, `make_figs.py` and `make_paper_assets.py` write `numbers.tex`, the table bodies and the figures. A one-command smoke test, `code/smoke_test.py`, runs the whole chain on a few stocks in about a minute and prints the accuracy next to the majority-class rate, so the pipeline can be confirmed before the full run is attempted.

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